BTS SIO SISR · SERVICES RÉSEAU LINUX

DHCP Failover sous Linux

Fiche procédure — serveurs DHCP ISC et KEA en haute disponibilité, relais DHCP sur routeur Linux, pfSense

Réalisé par	Thomas HIRON
Date	30/10/2026
Heure de début	30/10/2026 11:28
Heure de fin	30/10/2026 11:53
Poste	Pop!_OS 24.04 — KVM/libvirt (Lab SISR Manager)

Plan d'adressage

Réseau	Adresse	Machines
WAN	NAT (DHCP)	pfSense WAN
LAN-RT	172.20.20.0/29	pfSense LAN .2 — RTUX ens3 .6
LAN-SRV	172.20.10.0/28	SRV-DEB1 .1 — SRV-DEB2 .2 — RTUX ens4 .14
LAN-CLT	172.18.0.0/17	DEB-CLT (DHCP) — RTUX ens5 172.18.127.254

Domaine : **hiron.th** — DNS : **1.1.1.1** — Plage DHCP : 172.18.0.1 → 172.18.127.252

Sommaire

1. SRV-DEB1 — serveur DHCP principal
2. SRV-DEB2 — serveur DHCP de secours (KEA failover)
3. RTUX — routeur Linux et relais DHCP
4. pfSense — interfaces, filtrage et routes statiques
5. Client — DORA, identification du serveur et basculement
6. Synthèse

SRV-DEB1 — serveur DHCP principal

D'abord ISC DHCP comme demandé, puis KEA pour pouvoir former un failover avec SRV-DEB2.

1.1 — SRV-DEB1 — adresse IP statique (netplan)

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ hostname -f; grep -v '^#' /etc/netplan/50-cloud-init.yaml; echo; ip -4 -br addr; ip route
SRV-DEB1.hiron.th
network:
    ethernets:
        nic0:
            addresses:
            - 172.20.10.1/28
            gateway4: 172.20.10.14
            match:
                macaddress: 52:54:00:ab:9f:03
            nameservers:
                addresses:
                - 1.1.1.1
                search:
                - hiron.th
    version: 2

lo               UNKNOWN        127.0.0.1/8
ens3             UP             172.20.10.1/28
default via 172.20.10.14 dev ens3 proto static
172.20.10.0/28 dev ens3 proto kernel scope link src 172.20.10.1
```

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1.2 — SRV-DEB1 — installation du serveur DHCP ISC

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ dpkg -l isc-dhcp-server | tail -1; grep -v '^#' /etc/default/isc-dhcp-server | grep .; sy
stemctl status isc-dhcp-server --no-pager | head -5
ii  isc-dhcp-server 4.4.3-P1-2   amd64        ISC DHCP server for automatic IP address assignment
INTERFACESv4="ens3"
INTERFACESv6=""
● isc-dhcp-server.service - LSB: DHCP server
     Loaded: loaded (/etc/init.d/isc-dhcp-server; generated)
     Active: active (running) since Fri 2026-10-30 11:38:03 CET; 1min 48s ago
       Docs: man:systemd-sysv-generator(8)
    Process: 1204 ExecStart=/etc/init.d/isc-dhcp-server start (code=exited, status=0/SUCCESS)
```

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1.3 — SRV-DEB1 — fichier /etc/dhcp/dhcpd.conf (domaine hiron.th, DNS 1.1.1.1)

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ cat /etc/dhcp/dhcpd.conf; sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf 2>&1 | grep -i -E 'confi
g|error'
# SRV-DEB1 - Serveur DHCP ISC - TP DHCP Failover (domaine hiron.th)
option domain-name "hiron.th";
option domain-name-servers 1.1.1.1;
default-lease-time 600;
max-lease-time 7200;
authoritative;
log-facility local7;

# Reseau LAN-SRV (interface d ecoute du serveur, aucune distribution)
subnet 172.20.10.0 netmask 255.255.255.240 {
}

# Reseau LAN-CLT 172.18.0.0/17 (clients joints via le relais DHCP de RTUX)
subnet 172.18.0.0 netmask 255.255.128.0 {
  range 172.18.0.1 172.18.127.252;
  option routers 172.18.127.254;
  option broadcast-address 172.18.127.255;
  option subnet-mask 255.255.128.0;
}
Config file: /etc/dhcp/dhcpd.conf
```

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1.4 — SRV-DEB1 — premier bail ISC distribué via le relais de RTUX

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ sudo journalctl -u isc-dhcp-server --no-pager -o short -n 4 | cut -c1-200
oct. 30 11:39:31 SRV-DEB1 dhcpd[1216]: DHCPDISCOVER from 52:54:00:ab:05:61 via 172.18.127.254
oct. 30 11:39:32 SRV-DEB1 dhcpd[1216]: DHCPOFFER on 172.18.100.204 to 52:54:00:ab:05:61 (deb-clt) via 172.
18.127.254
oct. 30 11:39:32 SRV-DEB1 dhcpd[1216]: DHCPREQUEST for 172.18.100.204 (172.20.10.1) from 52:54:00:ab:05:61
(deb-clt) via 172.18.127.254
oct. 30 11:39:32 SRV-DEB1 dhcpd[1216]: DHCPACK on 172.18.100.204 to 52:54:00:ab:05:61 (deb-clt) via 172.1
8.127.254
```

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1.5 — SRV-DEB1 — passage d'ISC à KEA (nécessaire pour le failover avec SRV-DEB2)

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ sudo systemctl disable --now isc-dhcp-server
sudo systemctl restart kea-ctrl-agent kea-dhcp4-server
systemctl is-active isc-dhcp-server kea-dhcp4-server kea-ctrl-agent
grep -n this-server-name /etc/kea/kea-dhcp4.conf ; grep http-host /etc/kea/kea-ctrl-agent.conf
sudo ss -lunp | grep ':67 '
curl -X POST -d '{"command":"ha-heartbeat","service":["dhcp4"]}' http://172.20.10.1:8000/
isc-dhcp-server : inactive
kea-dhcp4-server / kea-ctrl-agent : active active
42:                    "this-server-name": "srv-deb1",
    "http-host": "172.20.10.1",
172.20.10.1:67 users:(("kea-dhcp4",pid=1988,fd=17))
Etat HA : hot-standby - portee : ['srv-deb1']
```

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SRV-DEB2 — serveur DHCP de secours (KEA failover)

Le module de haute disponibilité de KEA synchronise chaque bail entre les deux serveurs.

2.1 — SRV-DEB2 — adresse IP statique (netplan)

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ hostname -f; grep -v '^#' /etc/netplan/50-cloud-init.yaml; echo; ip -4 -br addr; ip route
SRV-DEB2.hiron.th
network:
    ethernets:
        nic0:
            addresses:
            - 172.20.10.2/28
            gateway4: 172.20.10.14
            match:
                macaddress: 52:54:00:ab:69:be
            nameservers:
                addresses:
                - 1.1.1.1
                search:
                - hiron.th
    version: 2

lo               UNKNOWN        127.0.0.1/8
ens3             UP             172.20.10.2/28
default via 172.20.10.14 dev ens3 proto static
172.20.10.0/28 dev ens3 proto kernel scope link src 172.20.10.2
```

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2.2 — SRV-DEB2 — installation de KEA (serveur DHCPv4 + agent de contrôle)

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ dpkg -l kea-dhcp4-server kea-ctrl-agent | tail -2 | cut -c1-90; ls /usr/lib/x86_64-linux-
gnu/kea/hooks/ | grep -E 'ha|lease_cmds'; systemctl is-active kea-dhcp4-server kea-ctrl-agent
ii  kea-ctrl-agent   2.2.0-6      amd64        REST API service for Kea DHCP server
ii  kea-dhcp4-server 2.2.0-6      amd64        IPv4 DHCP server
libdhcp_ha.so
libdhcp_lease_cmds.so
active
active
```

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2.3 — SRV-DEB2 — /etc/kea/kea-dhcp4.conf (1/2) : écoute, baux, options domaine et DNS, plage LAN-CLT

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ sed -n '1,31p;67,88p' /etc/kea/kea-dhcp4.conf     # début et fin du fichier
// SRV-DEB2 - Serveur DHCP KEA en haute disponibilite (failover) - TP DHCP Failover - domaine hiron.th
{
"Dhcp4": {
    // Ecoute sur la carte du reseau LAN-SRV (les clients arrivent via le relais de RTUX)
    "interfaces-config": {
        "interfaces": [ "ens3" ]
    },

    // Socket de commande utilise par l'agent de controle (echanges HA entre les 2 serveurs)
    "control-socket": {
        "socket-type": "unix",
        "socket-name": "/run/kea/kea4-ctrl-socket"
    },

    // Base des baux
    "lease-database": {
        "type": "memfile",
        "persist": true,
        "name": "/var/lib/kea/kea-leases4.csv",
        "lfc-interval": 3600
    },

    "valid-lifetime": 600,
    "max-valid-lifetime": 7200,

    // Options communes : domaine et DNS
    "option-data": [
        { "name": "domain-name", "data": "hiron.th" },
        { "name": "domain-name-servers", "data": "1.1.1.1" }
    ],

    // Reseau LAN-CLT 172.18.0.0/17
    "subnet4": [
        {
            "id": 1,
            "subnet": "172.18.0.0/17",
            "pools": [ { "pool": "172.18.0.1 - 172.18.127.252" } ],
            "option-data": [
                { "name": "routers", "data": "172.18.127.254" },
                { "name": "broadcast-address", "data": "172.18.127.255" }
            ]
        }
    ],

    "loggers": [
        {
            "name": "kea-dhcp4",
            "output_options": [ { "output": "syslog" } ],
            "severity": "INFO"
        }
    ]
}
}
```

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2.3b — SRV-DEB2 — /etc/kea/kea-dhcp4.conf (2/2) : failover KEA (module HA, hot-standby)

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ sed -n '32,66p' /etc/kea/kea-dhcp4.conf     # section haute disponibilité
    // Haute disponibilite : SRV-DEB1 principal, SRV-DEB2 secours (hot-standby)
    "hooks-libraries": [
        {
            "library": "/usr/lib/x86_64-linux-gnu/kea/hooks/libdhcp_lease_cmds.so",
            "parameters": { }
        },
        {
            "library": "/usr/lib/x86_64-linux-gnu/kea/hooks/libdhcp_ha.so",
            "parameters": {
                "high-availability": [ {
                    "this-server-name": "srv-deb2",
                    "mode": "hot-standby",
                    "heartbeat-delay": 5000,
                    "max-response-delay": 15000,
                    "max-ack-delay": 5000,
                    "max-unacked-clients": 0,
                    "peers": [
                        {
                            "name": "srv-deb1",
                            "url": "http://172.20.10.1:8000/",
                            "role": "primary",
                            "auto-failover": true
                        },
                        {
                            "name": "srv-deb2",
                            "url": "http://172.20.10.2:8000/",
                            "role": "standby",
                            "auto-failover": true
                        }
                    ]
                } ]
            }
        }
    ],
```

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2.4 — SRV-DEB2 — agent de contrôle KEA (canal HA, port 8000)

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ grep -E 'http-host|http-port' /etc/kea/kea-ctrl-agent.conf; ss -ltnp 2>/dev/null | grep 8
000 | cut -c1-80; sudo kea-dhcp4 -t /etc/kea/kea-dhcp4.conf | tail -1
    "http-host": "172.20.10.2",
    "http-port": 8000,
LISTEN 0      4096     172.20.10.2:8000      0.0.0.0:*
2026-10-30 11:41:56.251 INFO  [kea-dhcp4.dhcpsrv/1342.139861728432640] DHCPSRV_CFGMGR_NEW_SUBNET4 a new su
bnet has been added to configuration: 172.18.0.0/17 with params: valid-lifetime=600
```

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2.5 — SRV-DEB2 — état du failover : les deux serveurs en HOT-STANDBY

sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ curl -X POST -d '{"command":"ha-heartbeat","service":["dhcp4"]}' http://172.20.10.1:8000/
# interroge le partenaire
journalctl -u kea-dhcp4-server | grep HA_STATE
[
    {
        "arguments": {
            "date-time": "Fri, 30 Oct 2026 10:41:56 GMT",
            "scopes": [
                "srv-deb1"
            ],
            "state": "hot-standby",
            "unsent-update-count": 0
        },
        "result": 0,
        "text": "HA peer status returned."
    }
]
INFO  [kea-dhcp4.ha-hooks.139901897167360] HA_STATE_TRANSITION server transitions from SYNCING to READY st
ate, partner state is READY
INFO  [kea-dhcp4.ha-hooks.139901897167360] HA_STATE_TRANSITION server transitions from READY to HOT-STANDB
Y state, partner state is HOT-STANDBY
INFO  [kea-dhcp4.ha-hooks.139901897167360] HA_LOCAL_DHCP_ENABLE local DHCP service is enabled while the sr
v-deb2 is in the HOT-STANDBY state
```

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RTUX — routeur Linux et relais DHCP

Les clients ne sont pas sur le réseau des serveurs : le broadcast DHCP doit être relayé.

3.1 — RTUX — 3 adresses IP statiques (LAN-RT, LAN-SRV, LAN-CLT)

sisr@RTUX

```
sisr@RTUX:~$ sudo cat /etc/netplan/60-lab.yaml
ip -4 -br addr
network:
  version: 2
  ethernets:
    lab0:
      match: {macaddress: "52:54:00:ab:a1:ae"}
      addresses: [172.20.20.6/29]
      routes: [{to: default, via: 172.20.20.2}]
      nameservers: {addresses: [1.1.1.1], search: [hiron.th]}
    lab1:
      match: {macaddress: "52:54:00:ab:47:93"}
      addresses: [172.20.10.14/28]
    lab2:
      match: {macaddress: "52:54:00:ab:e7:eb"}
      addresses: [172.18.127.254/17]

lo               UNKNOWN        127.0.0.1/8
ens3             UP             172.20.20.6/29
ens4             UP             172.20.10.14/28
ens5             UP             172.18.127.254/17
```

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3.1b — RTUX — passerelle par défaut = pfSense (172.20.20.2)

sisr@RTUX

```
sisr@RTUX:~$ ip route
ping -c2 172.20.20.2
ping -c2 1.1.1.1
default via 172.20.20.2 dev ens3 proto static
172.18.0.0/17 dev ens5 proto kernel scope link src 172.18.127.254
172.20.10.0/28 dev ens4 proto kernel scope link src 172.20.10.14
172.20.20.0/29 dev ens3 proto kernel scope link src 172.20.20.6

2 packets transmitted, 2 received, 0% packet loss, time 1023ms
rtt min/avg/max/mdev = 0.289/0.293/0.298/0.004 ms

2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 12.142/13.077/14.012/0.935 ms
```

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3.2 — RTUX — activation du routage

sisr@RTUX

```
sisr@RTUX:~$ cat /etc/sysctl.d/99-routage.conf; /sbin/sysctl net.ipv4.ip_forward
net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
```

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3.3 — RTUX — informations générales : domaine hiron.th et DNS 1.1.1.1

sisr@RTUX

```
sisr@RTUX:~$ hostname -f; resolvectl dns ens3; resolvectl domain ens3; getent ahostsv4 deb.debian.org | he
ad -1
RTUX.hiron.th
Link 2 (ens3): 1.1.1.1
Link 2 (ens3): hiron.th
151.101.2.132   STREAM debian.map.fastlydns.net
```

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3.4 — RTUX — relais DHCP vers SRV-DEB1 et SRV-DEB2

sisr@RTUX

```
sisr@RTUX:~$ grep -v '^#' /etc/default/isc-dhcp-relay | grep .; systemctl status isc-dhcp-relay --no-pager
| head -4
SERVERS="172.20.10.1 172.20.10.2"
INTERFACES="ens4 ens5"
OPTIONS=""
● isc-dhcp-relay.service - LSB: DHCP relay
     Loaded: loaded (/etc/init.d/isc-dhcp-relay; generated)
     Active: active (running) since Fri 2026-10-30 11:39:18 CET; 33s ago
       Docs: man:systemd-sysv-generator(8)
```

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pfSense — interfaces, filtrage et routes statiques

pfSense ne connaît que LAN-RT : il faut lui indiquer que LAN-SRV et LAN-CLT sont derrière RTUX.

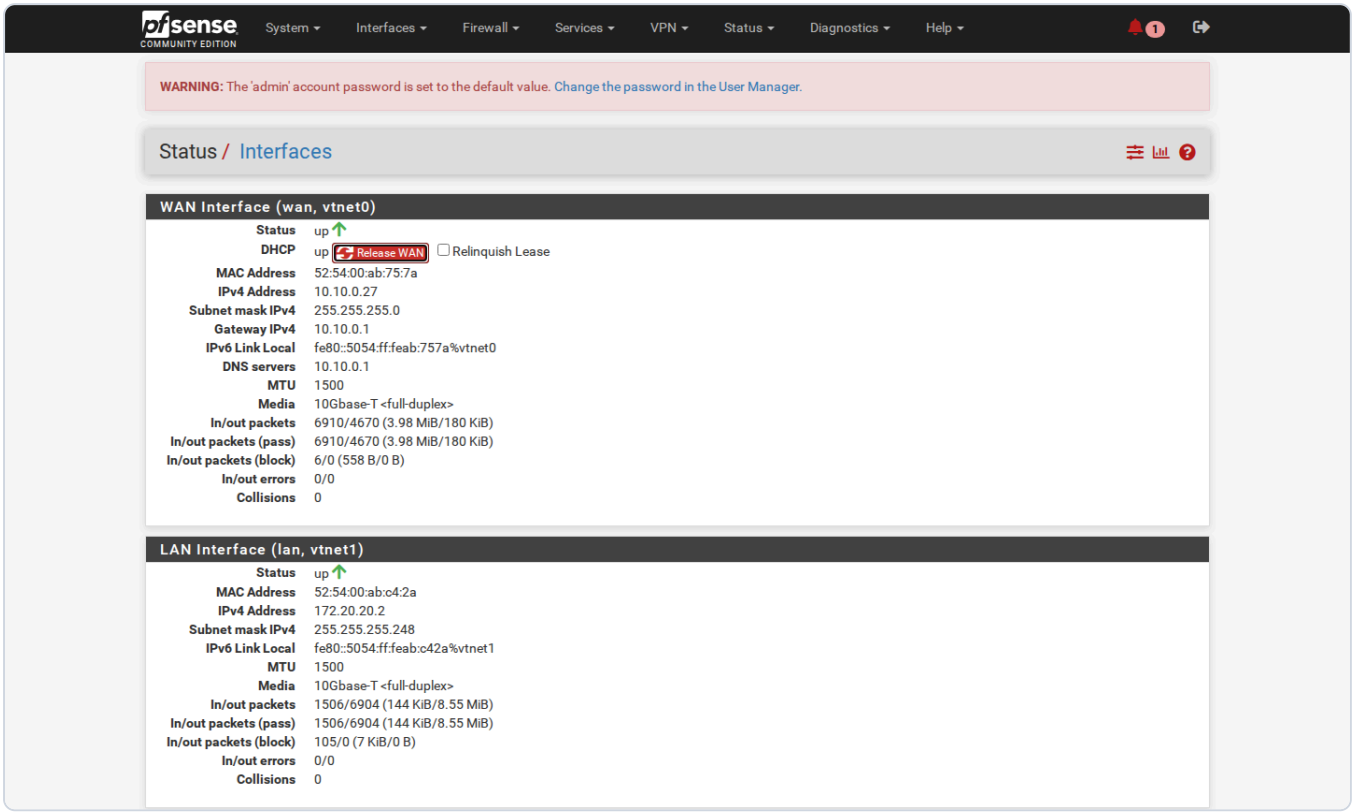
4.0 — DEB-CLT — le client joint l'interface web de pfSense (via RTUX) et Internet

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ curl -k https://172.20.20.2/ | grep title      # interface web de pfSense
ping -c2 172.20.20.2 ; ping -c2 1.1.1.1
curl -I http://deb.debian.org/debian/
sudo traceroute -I -n -m 4 1.1.1.1
<title>pfSense - Login</title>
rtt min/avg/max/mdev = 0.588/1.033/1.479/0.445 ms
rtt min/avg/max/mdev = 15.798/15.939/16.081/0.141 ms
HTTP/1.1 200 OK
traceroute to 1.1.1.1 (1.1.1.1), 4 hops max, 60 byte packets
 1  172.18.127.254  0.869 ms  0.779 ms  0.761 ms
 2  172.20.20.2  1.795 ms  1.783 ms  1.771 ms
 3  10.10.0.1  2.484 ms  2.475 ms  2.465 ms
 4  10.101.128.1  8.152 ms  8.141 ms  7.888 ms
```

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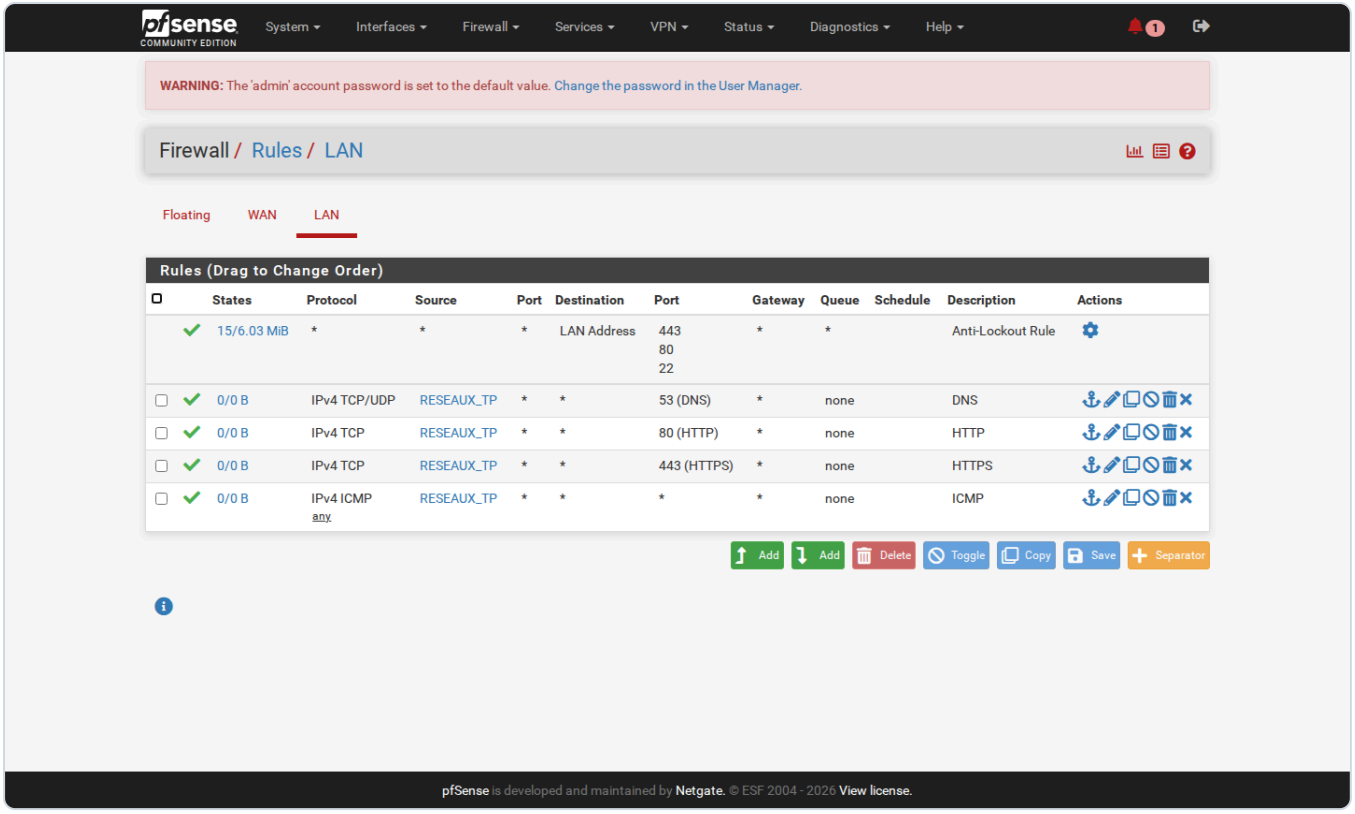
4.1 — pfSense — Interfaces réseau (WAN / LAN)



Source : https://172.20.20.2/status_interfaces.php

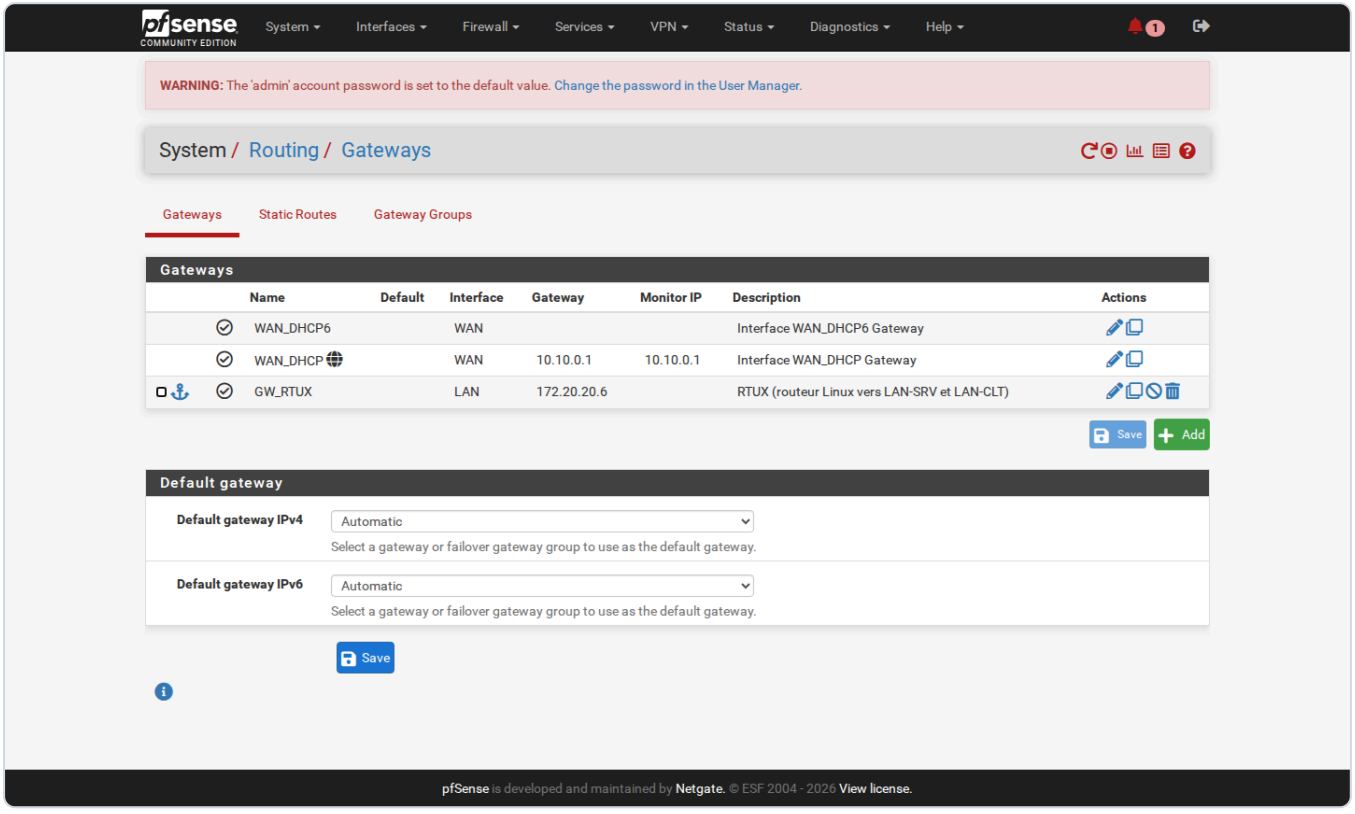
Capturé le 30/10/2026 11:37:10

4.2 — pfSense — Règles du pare-feu LAN : DNS, HTTP/S et ICMP uniquement



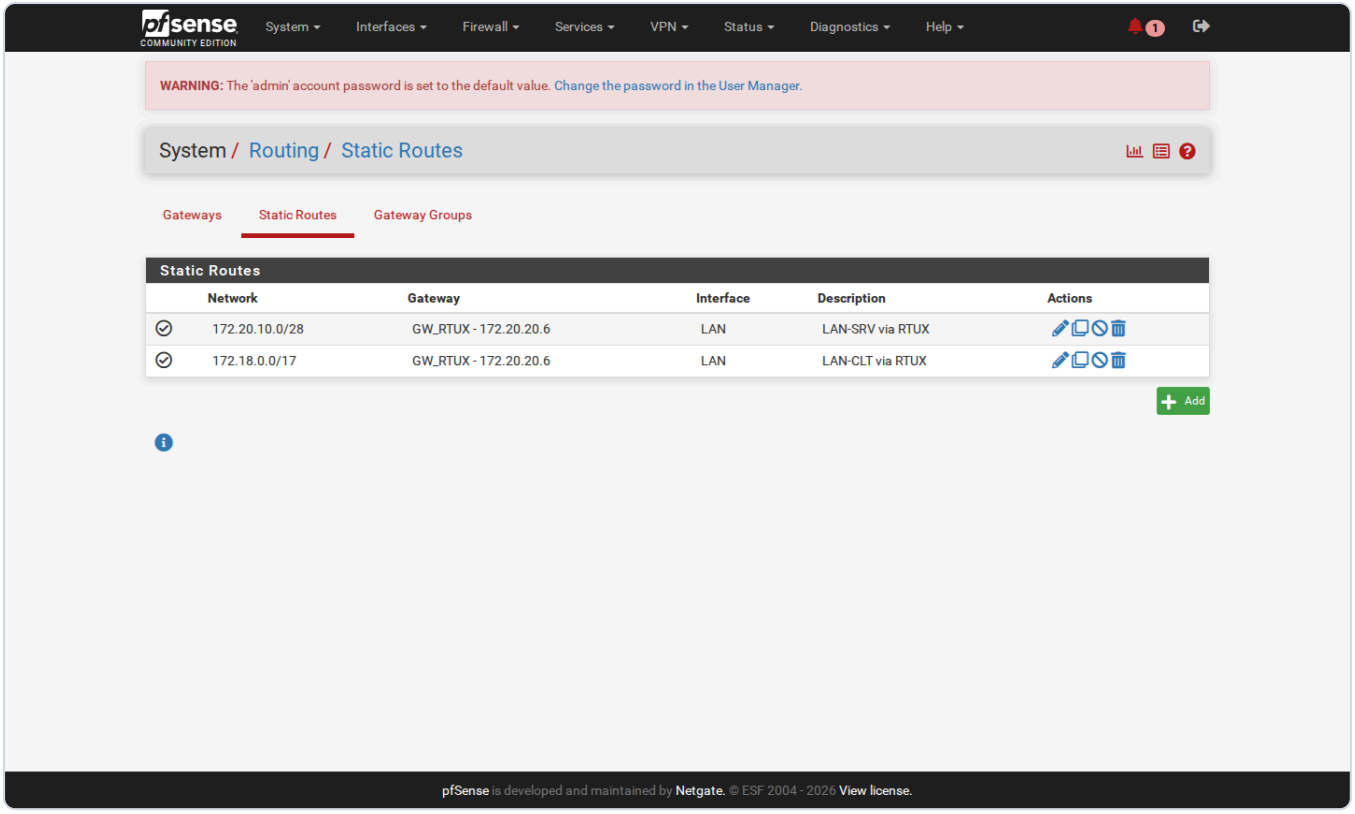
Source : https://172.20.20.2/firewall_rules.php?if=lan
Capturé le 30/10/2026 11:37:17

4.3 — pfSense — Passerelle GW_RTUX



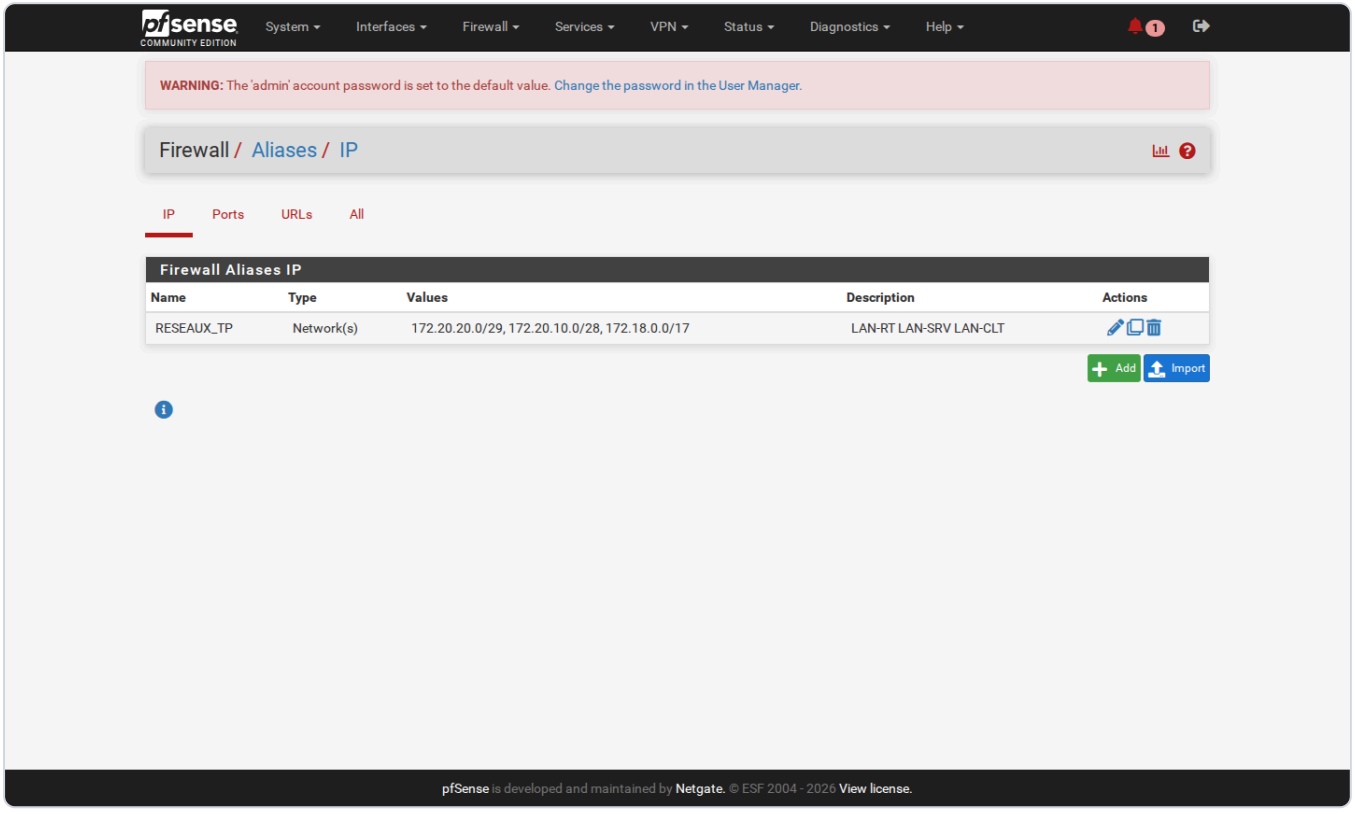
Source : https://172.20.20.2/system_gateways.php
Capturé le 30/10/2026 11:37:23

4.4 — pfSense — Routes statiques vers LAN-SRV et LAN-CLT



Source : https://172.20.20.2/system_routes.php
Capturé le 30/10/2026 11:37:30

4.5 — pfSense — Alias RESEAUX_TP



Source : https://172.20.20.2/firewall_aliases.php
Capturé le 30/10/2026 11:37:37

Client — DORA, identification du serveur et basculement

On observe l'échange DHCP, on identifie le serveur qui répond puis on l'arrête.

5.1 — DEB-CLT — Wireshark installé (accès à la capture pour l'utilisateur)

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ sudo apt install wireshark tshark      # réponse « Oui » : les non-superutilisateurs peuve
nt capturer
sudo usermod -aG wireshark sisr
dpkg -l wireshark tshark ; id sisr
ii  tshark         4.0.17-0+deb12u3 amd64        network traffic analyzer -
ii  wireshark      4.0.17-0+deb12u3 amd64        network traffic analyzer -
uid=1001(sisr) gid=1001(sisr) groupes=1001(sisr),27(sudo),114(wireshark)
```

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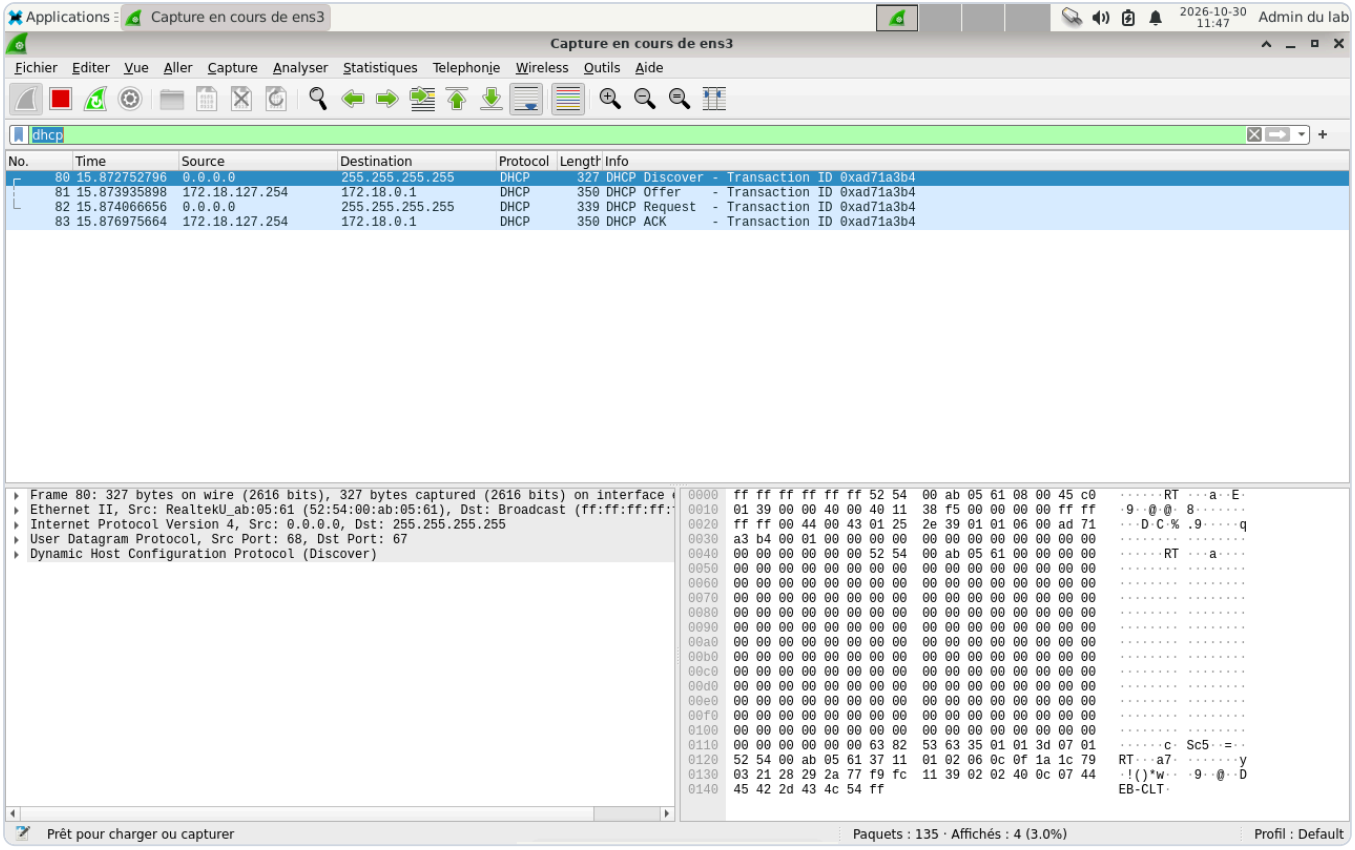
5.2 — DEB-CLT — interface en adressage automatique (DHCP) avec NetworkManager

sisr@DEB-CLT

```
sisr@DEB-CLT:~$ nmcli -f NAME,DEVICE,TYPE con show --active
sudo nmcli con modify 'Connexion filaire 1' ipv4.method auto     # « Automatique (DHCP) »
nmcli -g ipv4.method con show 'Connexion filaire 1'
ip -4 -br addr show ens3 ; ip route
NAME                 DEVICE  TYPE
Connexion filaire 1  ens3    ethernet
lo                   lo      loopback
auto
ens3             UP             172.18.0.1/17
default via 172.18.127.254 dev ens3 proto dhcp src 172.18.0.1 metric 100
172.18.0.0/17 dev ens3 proto kernel scope link src 172.18.0.1 metric 100
```

Capturé le 30/10/2026 11:50:08

5.3 — DEB-CLT — Wireshark filtré sur DHCP : échange DORA



Source : écran de la VM lab-deb-clt

Capturé le 30/10/2026 11:48:00

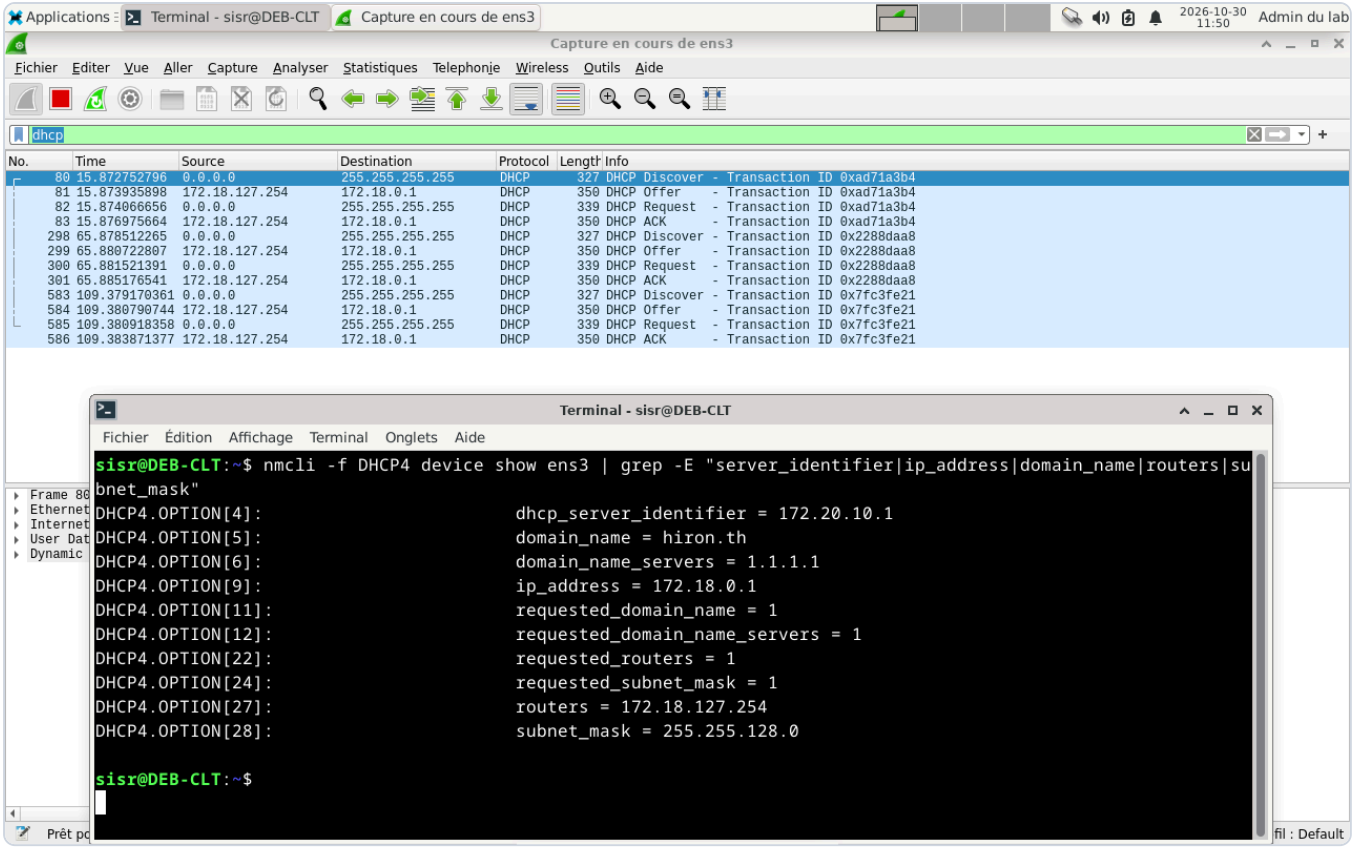
5.3b — DEB-CLT — détail des 4 messages DORA (tshark, version texte de Wireshark)

```
sisr@DEB-CLT
sisr@DEB-CLT:~$ tshark -r dora1.pcapng -Y dhcp -T fields -e frame.number -e ip.src -e ip.dst -e dhcp.optio
n.dhcp -e dhcp.ip.your -e dhcp.ip.relay -e dhcp.option.dhcp_server_id | column -t
tshark -r dora1.pcapng -Y 'dhcp.option.dhcp == 5' -O dhcp | grep -E 'Your|Relay|Server Identifier|Subnet M
ask|Router|Domain|Lease'
frame.number  ip.src          ip.dst          dhcp.option.dhcp  dhcp.ip.your  dhcp.ip.relay   dhcp.optio
n.dhcp_server_id
1             0.0.0.0         255.255.255.255  1                0.0.0.0       0.0.0.0
2             172.18.127.254  172.18.0.1       2                172.18.0.1    172.18.127.254  172.20.10.1
3             0.0.0.0         255.255.255.255  3                0.0.0.0       0.0.0.0         172.20.10.1
4             172.18.127.254  172.18.0.1       5                172.18.0.1    172.18.127.254  172.20.10.1

    Your (client) IP address: 172.18.0.1
    Relay agent IP address: 172.18.127.254
        Subnet Mask: 255.255.128.0
        Router: 172.18.127.254
        Domain Name Server: 1.1.1.1
        Domain Name: hiron.th
        IP Address Lease Time: (600s) 10 minutes
    Option: (54) DHCP Server Identifier (172.20.10.1)
        DHCP Server Identifier: 172.20.10.1
```

Capturé le 30/10/2026 11:49:52

5.4 — DEB-CLT — nmcli : le bail a été fourni par SRV-DEB1 (172.20.10.1)



Source : écran de la VM lab-deb-clt

Capturé le 30/10/2026 11:50:23

5.5 — SRV-DEB1 — arrêt du serveur DHCP principal

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ sudo systemctl stop kea-dhcp4-server
systemctl is-active kea-dhcp4-server
date '+%H:%M:%S'
inactive
11:50:34
```

Capturé le 30/10/2026 11:50:35

5.6 — SRV-DEB2 — le secours détecte la panne et passe en PARTNER-DOWN

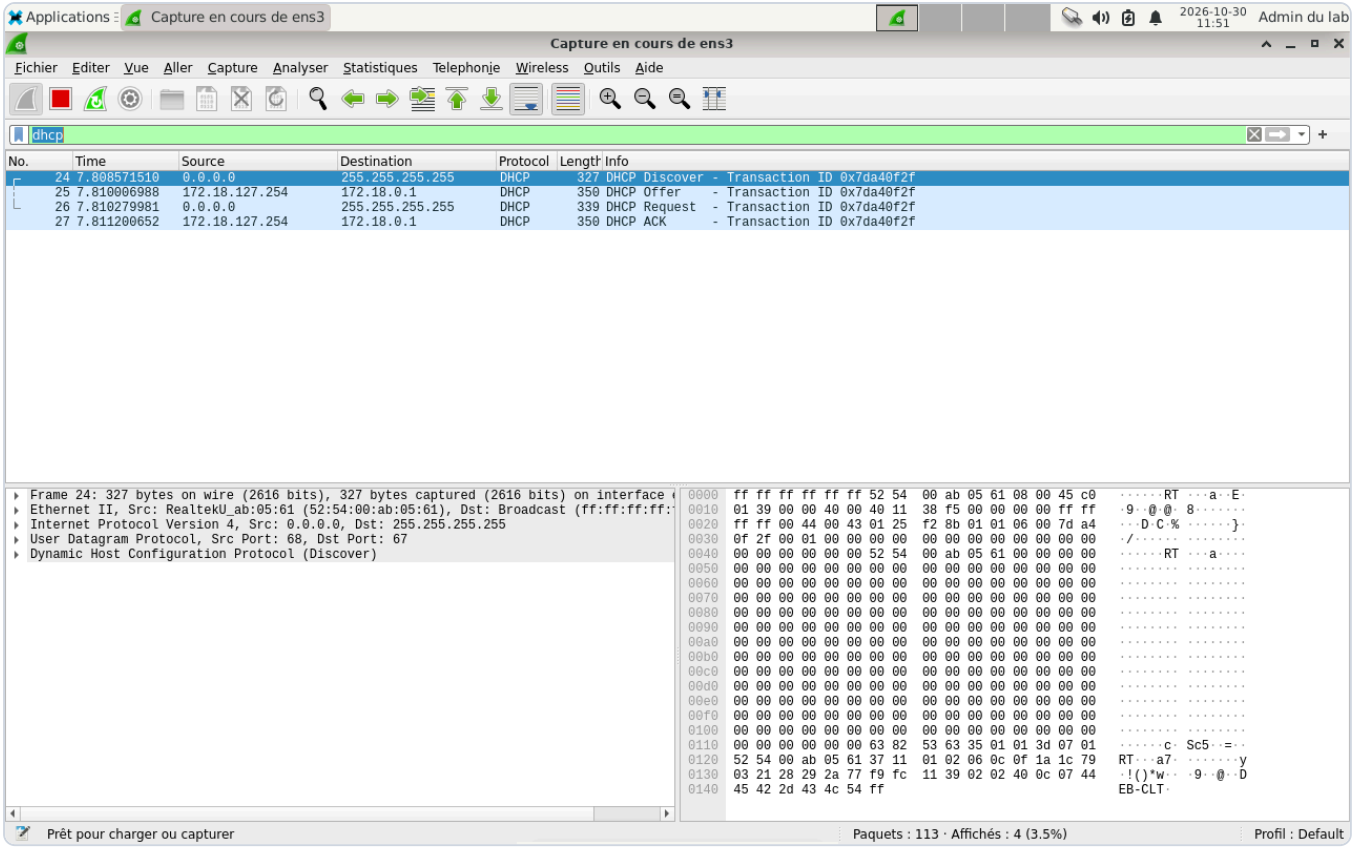
sisr@SRV-DEB2

```
sisr@SRV-DEB2:~$ journalctl -u kea-dhcp4-server --since '-3min' | grep HA_
curl -X POST -d '{"command":"ha-heartbeat","service":["dhcp4"]}' http://172.20.10.2:8000/
oct. 30 11:50:49 SRV-DEB2 WARN  HA_COMMUNICATION_INTERRUPTED communication with srv-deb1 is interrupted
oct. 30 11:50:49 SRV-DEB2 INFO  HA_STATE_TRANSITION server transitions from HOT-STANDBY to PARTNER-DOWN st
ate, partner state is UNDEFINED
oct. 30 11:50:49 SRV-DEB2 INFO  HA_LEASE_UPDATES_DISABLED lease updates will not be sent to the partner wh
ile in PARTNER-DOWN state

Etat HA de SRV-DEB2 : partner-down - portee servie : ['srv-deb1']
```

Capturé le 30/10/2026 11:50:50

5.7 — DEB-CLT — Wireshark après l'arrêt de SRV-DEB1 : nouveau DORA, servi par SRV-DEB2



Source : écran de la VM lab-deb-clt

Capturé le 30/10/2026 11:51:16

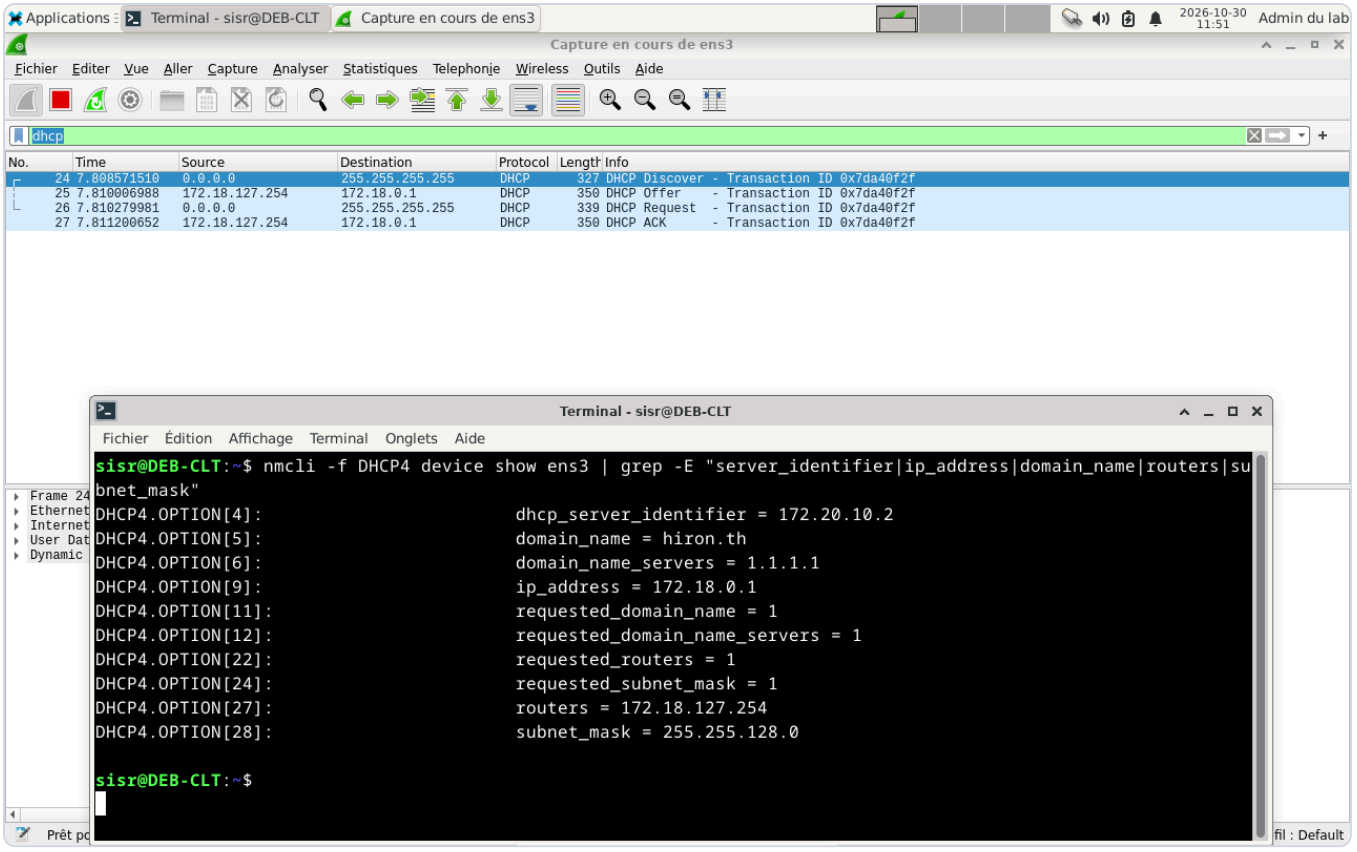
5.7b — DEB-CLT — DORA après basculement : DHCP Server Identifier = 172.20.10.2

```
sisr@DEB-CLT

sisr@DEB-CLT:~$ tshark -r dora2.pcapng -Y dhcp -T fields -e frame.number -e ip.src -e ip.dst -e dhcp.optio
n.dhcp -e dhcp.ip.your -e dhcp.ip.relay -e dhcp.option.dhcp_server_id | column -t
frame.number  ip.src          ip.dst          dhcp.option.dhcp  dhcp.ip.your  dhcp.ip.relay   dhcp.optio
n.dhcp_server_id
1             0.0.0.0         255.255.255.255  1                0.0.0.0       0.0.0.0
2             172.18.127.254  172.18.0.1       2                172.18.0.1    172.18.127.254  172.20.10.2
3             0.0.0.0         255.255.255.255  3                0.0.0.0       0.0.0.0         172.20.10.2
4             172.18.127.254  172.18.0.1       5                172.18.0.1    172.18.127.254  172.20.10.2
```

Capturé le 30/10/2026 11:51:29

5.8 — DEB-CLT — nmcli : le bail est maintenant fourni par SRV-DEB2 (172.20.10.2), même adresse IP



Source : écran de la VM lab-deb-clt
Capturé le 30/10/2026 11:51:20

5.9 — SRV-DEB1 — redémarrage : resynchronisation des baux puis retour en HOT-STANDBY

sisr@SRV-DEB1

```
sisr@SRV-DEB1:~$ sudo systemctl start kea-dhcp4-server
journalctl -u kea-dhcp4-server --since '-40s' | grep -E 'HA_STATE_TRANSITION|SYNC'
oct. 30 11:51:35 SRV-DEB1 INFO  HA_STATE_TRANSITION server transitions from WAITING to SYNCING state, partner state is PARTNER-DOWN
oct. 30 11:51:35 SRV-DEB1 INFO  HA_SYNC_START starting lease database synchronization with srv-deb2
oct. 30 11:51:35 SRV-DEB1 INFO  HA_LEASES_SYNC_LEASE_PAGE_RECEIVED received 1 leases from srv-deb2
oct. 30 11:51:35 SRV-DEB1 INFO  HA_SYNC_SUCCESSFUL lease database synchronization with srv-deb2 completed successfully in 5.603 ms
oct. 30 11:51:35 SRV-DEB1 INFO  HA_STATE_TRANSITION server transitions from SYNCING to READY state, partner state is PARTNER-DOWN
oct. 30 11:51:45 SRV-DEB1 INFO  HA_STATE_TRANSITION server transitions from READY to HOT-STANDBY state, partner state is HOT-STANDBY
```

Capturé le 30/10/2026 11:51:55

Synthèse

6.1 — Ce qu'il faut retenir

1. Un serveur DHCP placé sur un autre réseau que ses clients a besoin d'un relais (isc-dhcp-relay sur RTUX) : le relais renseigne le champ giaddr (172.18.127.254), ce qui permet au serveur de choisir la bonne plage. 2. Le masque doit être identique partout : avec un /28 sur RTUX, le relais aurait annoncé un réseau 172.18.127.240/28 incompatible avec la plage 172.18.0.0/17 des clients. 3. ISC DHCP et KEA ne peuvent pas être partenaires de failover : le failover a été réalisé en KEA ↔ KEA (module libdhcp_ha.so, mode hot-standby, échanges via kea-ctrl-agent sur le port 8000). 4. Le relais pointe vers les deux serveurs. 5. Test : arrêt du principal → le secours passe PARTNER-DOWN en 15 s et sert le client avec la même adresse (baux répliqués) ; au retour du principal, synchronisation des baux puis retour en HOT-STANDBY. 6. pfSense doit connaître les réseaux situés derrière RTUX (routes statiques via la passerelle GW_RTUX) ; son NAT sortant automatique les inclut alors, et le filtrage LAN se limite à DNS, HTTP/S et ICMP.

Capturé le 30/10/2026 11:53:22